Week 8

Spring 2027 · Section 017 · MW 7:30–8:50 PM · RRC Room 8210

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a bootstrap sample and a bootstrap distribution are created. (CLO 5, CLO 6) **2.** Explain why a bootstrap distribution is centered at the sample statistic. (CLO 5) **3.** Use StatKey to generate a bootstrap distribution. (CLO 5, CLO 6) **4.** Estimate the standard error as the standard deviation of the bootstrap distribution. (CLO 5, CLO 6) **5.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap Confidence Intervals Your Turn: Bootstrap a Proportion (32 of 50 prefer Brand A): build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Mar 14, 11:59 PM) Take-Home Quiz 5 (due Mon Mar 22, 11:59 PM) – covers 3.3–3.4 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS Blackboard StatKey Respondus
6. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval. (CLO 6) **7.** Construct a percentile bootstrap confidence interval with StatKey. (CLO 6) **8.** Predict how changing the confidence level and sample size affect the width of an interval. (CLO 6) **9.** Evaluate whether a bootstrap distribution is symmetric and bell-shaped enough to trust a bootstrap interval. (CLO 6) **10.** Explain why a bootstrap interval cannot correct for a non-representative sample. (CLO 1, CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Confidence Intervals Using Percentiles – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods Your Turn: Percentile CIs (124 of 200 students work)	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Mar 14, 11:59 PM) Take-Home Quiz 5 (due Mon Mar 22, 11:59 PM) – covers 3.3–3.4 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus

Also this week: Spring Break – Mon Mar 15 – Sun Mar 21 (no class; college closed)